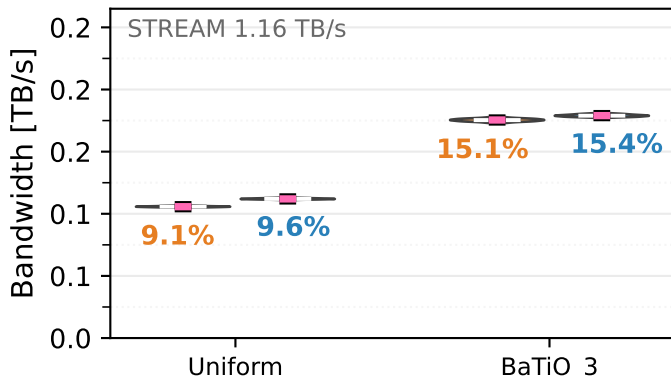


Gather-Accumulate-Scatter

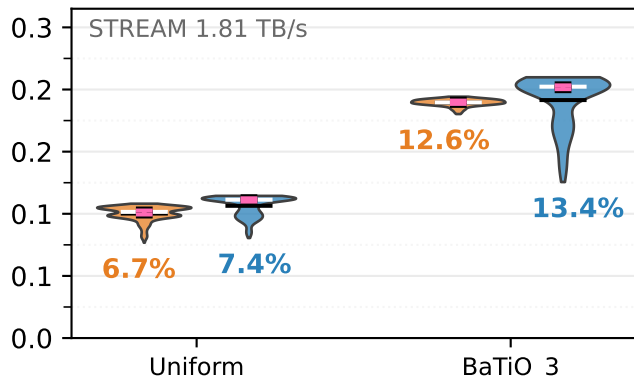
% annotations relative to STREAM peak bandwidth

Original: $y[\sigma(i)] += x[i]$; Shuffled: $y[\sigma(\pi(i))] += x[\pi(i)]$, $\pi = \text{argsort}(\sigma)$

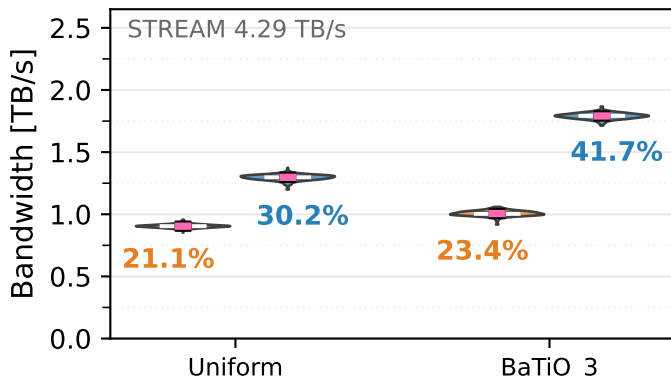
MI300A Zen CPU



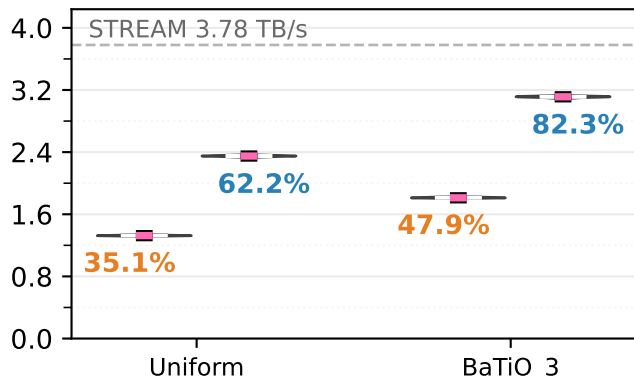
GH200 Grace CPU



MI300A GPU



GH200 Hopper GPU



Original Order Sorted (Shuffle)